BOTANORA

N° 04



Focus & Energy

Green tea, matcha and mate for energy without the crash

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Green tea, matcha and mate for energy without the crash

Good energy is steady energy. This guide shows how to use tea's natural pairing of caffeine and L-theanine for attention without spikes, how to space your cups through the day and how to swap the third coffee for something that does not send you the bill at night.

It includes a caffeine calculator to fill in, a three-cup routine and eighteen recipes for home and work, a weekly prep plan.



01

Know your limit

Add up your daily caffeine with the calculator.

02

Space your cups

Three moments, three teas, one cut-off time.

03

Swap gradually

A two-week plan avoids withdrawal headaches.

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CHAPTER I

Caffeine and L-theanine

The chemistry of calm alertness

Tea is the only common drink that carries, in the same leaf, a stimulant and an amino acid that softens its effects. Understanding that pair helps you pick the right tea for each hour.

01 How it works
02 How much caffeine each tea has
03 Daily calculator



CHAPTER I

How it works

Caffeine

The accelerator

Through the day the brain builds up adenosine, a molecule that signals tiredness. Caffeine takes its place on the receptors and postpones the feeling of sleepiness.

The effect starts in 20 to 45 minutes and the average half-life is about five hours, varying widely between people.

L-theanine

The damper

An amino acid almost unique to the tea plant, responsible for part of its sweetness and umami. It is associated with a relaxed, alert state.

It is most abundant in young, shaded teas such as gyokuro and matcha.

What the studies say

Controlled trials with caffeine and L-theanine together show modest improvements in attention tasks and reaction time, with less feeling of tension than caffeine alone. The effect is dose-dependent and clearest in people who do not consume a lot of caffeine.

In practice: tea gives a gentler rise and a softer landing than coffee. It is not magic, it is dose and company.

Coffee is a push. Tea is a current.

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Drinking tea with food and water alongside makes the energy curve even steadier.

CHAPTER I

How much caffeine each tea has

Ranges per 200 ml cup. Brewing matters as much as the type of tea.

DRINK	CAFFEINE	L-THEANINE	ENERGY PROFILE
Matcha (2 g)	40–70 mg	very high	Strong and long, yet smooth
Gyokuro	30–60 mg	very high	Calm and concentrated
Black tea	40–70 mg	medium	Direct, for mornings
Yerba mate	30–80 mg	none	Social, prolonged
Oolong	30–50 mg	medium	Balanced
Sencha	20–45 mg	high	Light and alert
White tea	15–35 mg	high	Subtle
Hojicha · genmaicha	10–25 mg	low	Late afternoon
Drip coffee (reference)	80–120 mg	none	Fast and intense

What changes the caffeine in your cup

- **Time and temperature:** the hotter and longer, the more caffeine extracted.
- **Amount of leaf:** doubling the dose nearly doubles the caffeine.
- **Buds and young leaves** contain more caffeine than mature leaves.
- **Broken leaf and bags** release faster than whole leaf.
- **Resteeps:** the first steep gives much of the caffeine; later ones less.

> COMMON MYTH
> Rinsing tea for 30 seconds does not "remove the caffeine". Half a minute extracts little; to really cut down, choose low-caffeine teas or tisanes.

TO PRINT

Daily calculator

For healthy adults, health authorities consider up to about 400 mg of caffeine a day safe; in pregnancy, up to about 200 mg. Add up a typical day.

DRINK	MG (AVERAGE)	HOW MANY TODAY?	TOTAL
Drip coffee (200 ml)	95		
Espresso (1 shot)	65		
Black tea (200 ml)	50		
Matcha (2 g)	60		
Green tea (200 ml)	30		
Oolong (200 ml)	40		
Mate (gourd, session)	80–150		
Cola (350 ml)	35		
Energy drink (250 ml)	80		
Dark chocolate (30 g)	20		
Daily total			

RESULT	READING
up to 200 mg	Light. Room for another cup in the morning.
200 to 400 mg	Moderate. Watch your cut-off time.
over 400 mg	High. Use the swap plan in chapter II.

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Teenagers and more sensitive people should stay well below these limits. When in doubt, a rule of thumb is about 3 mg per kilo of body weight per day.

CHAPTER II

The three-cup routine

Energy spread out

Instead of one big dose in the morning and another mid-afternoon, three smaller, decreasing cups. Energy stays steady and sleep stays protected.

01 The routine

02 Two-week swap plan

03 Work and study rituals



CHAPTER II

The routine

MOMENT	SUGGESTED TEA	CAFFEINE	WHY
Cup 1 · 60 to 90 min after waking	Black, matcha or mate	high	Natural morning cortisol already helps at first; caffeine works better a bit later
Cup 2 · mid-morning	Green or oolong	medium	Keeps focus through peak work
Cup 3 · before your cut-off	White, genmaicha or hojicha	low	Avoids the afternoon slump without a night-time cost
After that	Tisanes and water	none	Hydration and ritual

Your cut-off time

Count back six to eight hours from bedtime. If you sleep at 11 pm, stop caffeine between 3 and 5 pm. If you wake up tired even after eight hours, move the limit earlier and watch for a week.

Three rules

1. **Water alongside.** One glass of water for every cup.
2. **Food nearby.** Strong tea on an empty stomach can cause nausea or heartburn in sensitive people.
3. **Iron and meals.** Tannins reduce absorption of plant-based iron; people with anaemia should drink tea between meals, not with them.

CHAPTER II

Two-week swap plan

Cutting caffeine all at once brings headaches and irritability. Swapping gradually does not.

DAYS	MORNING	MID-MORNING	AFTERNOON
1–3	Coffee as usual	Coffee	Coffee → black tea
4–6	Coffee	Coffee → matcha	Black tea → oolong
7–9	Smaller coffee or half-and-half	Matcha → green	Oolong → genmaicha
10–12	Black tea or matcha	Green	Hojicha or white
13–14	The three-cup routine		

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If a headache shows up, go back one step for two days and continue. Water and regular sleep shorten the adjustment.

Signs of too much

- Shaky hands, racing heart
- Anxiety or irritability for no reason
- Heartburn and stomach upset
- Trouble falling asleep
- Light sleep, waking tired
- Headaches on caffeine-free days

NOTE

Palpitations, chest pain or shortness of breath are not normal: seek care.

CHAPTER II

Work and study rituals

The 50-minute block

One cup per block

Brew the tea before you start, work 50 minutes without notifications and use the 10-minute break to resteep. The tea becomes the clock.

The desk glass

Grandpa-style green

Loose leaves in a tall glass, topped up through the morning. It hydrates, gives little caffeine per sip and asks for nothing.

The gongfu break

15 minutes of attention

Mid-afternoon, an oolong in short steeps. The ritual makes you stop, look and breathe: it is active rest.

The cold bottle

Cold brew in your bag

A litre of green tea or tereré made the night before. Light energy for long hot days.

Focus is not intensity. It is continuity.

CHAPTER III

Eighteen recipes

From morning to late afternoon

Recipes for every moment, from the first cup of the day to the last. All fit an ordinary kitchen or an office pantry.

01 Matcha Ginger Shot
02 Matcha Lassi
03 Ice-brewed Gyokuro
04 Green, Lime & Ginger
05 Afternoon White with Mint
06 Orange & Rosemary Tereré
07 Green Tea with Guaraná and Lime
08 Black Tea Mocha
09 Oolong with Charred Orange
10 Mid-afternoon Genmaicha
11 Hojicha Latte
12 Green Tea & Coconut Water



01 Matcha Ginger Shot

A concentrated sip to start the day.

Hot · 3 min · 1 shot · 60 ml · Caffeinated · Easy

INGREDIENTS

2 g	matcha
40 ml	water at 75 °C
1 tsp	fresh ginger juice (grated and squeezed)
1 tsp	honey

METHOD

1. Grate the ginger and squeeze it through a clean cloth or fine strainer.
2. Whisk the matcha with the water and honey until frothy.
3. Add the ginger juice and drink in two sips.

BARISTA TIP It is the fastest way to have matcha on a busy morning: under 3 minutes, no milk.

02 Matcha Lassi

Yoghurt, matcha and cardamom: a breakfast you drink.

Iced · 4 min · 1 glass · 300 ml · Caffeinated · With milk · Easy

INGREDIENTS

170 g	plain whole yoghurt
100 ml	iced water
2 g	matcha
1 tbsp	honey
1 pinch	ground cardamom

Finish. Chopped pistachios.

METHOD

1. Blend everything for 30 s.
2. Serve over ice.

BARISTA TIP The protein and fat in the yoghurt slow the caffeine down: energy without a spike.

03 Ice-brewed Gyokuro

Koridashi: ice melts over the leaves into a sweet broth.

Iced · 40 min · 2 small cups · Caffeinated · Intermediate

INGREDIENTS

6 g	gyokuro or shaded sencha
150 g	ice cubes from filtered water

METHOD

1. Put the leaves in a kyusu or small pot with a strainer.
2. Cover with the ice cubes.
3. Let it melt at room temperature, 30 to 40 min.
4. Serve in small cups. Then make a quick steep with water at 60 °C.

BARISTA TIP Ice extracts mostly amino acids (umami and L-theanine) and very little caffeine: the calmest, most concentrated tea there is.

04 Green, Lime & Ginger

Wakes you up without weighing you down.

Hot · 6 min · 1 cup · 250 ml · Caffeinated · Easy

INGREDIENTS

3 slices	ginger
250 ml	water
3 g	Chinese green tea
1 strip	lime peel

Finish. Lime peel twisted over the cup.

METHOD

1. Simmer the ginger in the water for 3 min.
2. Turn off and wait 2 min (≈ 80 °C).
3. Add the green tea and lime peel; 2 min.
4. Strain.

BARISTA TIP The ginger simmers; the green tea never does. Always in that order.

05 Afternoon White with Mint

Little caffeine, lots of freshness.

Hot · 5 min · 1 pot · 400 ml · Caffeinated · Easy

INGREDIENTS

5 g	Shou Mei or Bai Mu Dan
5 leaves	fresh mint
400 ml	water at 85 °C

METHOD

1. Steep the white tea for 3 min.
2. Add the mint in the last minute.
3. Strain. Make a second 4-minute steep with the same leaves.

BARISTA TIP Good after lunch, when you want focus without hurting your sleep.

06 Orange & Rosemary Tereré

Cold, citrusy and resinous.

Iced · 5 min + 1 h · 1 bottle · 500 ml · Caffeinated · Easy

INGREDIENTS

2 tbsp	tereré yerba mate
300 ml	fresh orange juice
200 ml	iced water
1 small sprig	rosemary

METHOD

1. Put everything in a bottle.
2. Fridge for 1 h.
3. Strain and drink the same day.

BARISTA TIP Rosemary is traditionally linked to memory, and its scent makes a great break.

07 Green Tea with Guaraná and Lime

The Brazilian pairing for long days.

Iced · 5 min · 1 glass · 300 ml · Caffeinated · Easy

INGREDIENTS

250 ml	cold green tea (cold brew or flash)
¼ tsp	guaraná powder
½	lime (juice)
1 tsp	honey

METHOD

1. Dissolve the guaraná and honey in the lime juice.
2. Add the cold green tea and stir well.
3. Serve over ice.

BARISTA TIP Guaraná powder is high in caffeine (¼ tsp ≈ 40 to 60 mg). Do not exceed this dose and avoid it in the afternoon.

08 Black Tea Mocha

Black tea, cocoa and milk: the stand-in for a café mocha.

Hot · 6 min · 1 mug · 250 ml · Caffeinated · With milk · Easy

INGREDIENTS

3 g	Assam black tea
100 ml	boiling water
1 tbsp	100% cocoa powder
150 ml	milk
2 tsp	brown sugar

METHOD

1. Steep the tea for 4 min and strain.
2. Mix the cocoa and sugar with a little tea into a paste.
3. Add the rest of the tea and the hot frothed milk.

BARISTA TIP Cocoa contains theobromine, a gentle stimulant: it adds to black tea without overdoing it.

09 Oolong with Charred Orange

Roasted and citrusy for the afternoon break.

Hot · 8 min · 1 pot · 400 ml · Caffeinated · Intermediate

INGREDIENTS

5 g	roasted oolong (Dong Ding, Da Hong Pao)
2	thick orange slices
400 ml	water at 95 °C

METHOD

1. Char the orange slices in a very hot dry pan, 1 min each side.
2. Put them in the pot with the oolong.
3. Steep 3 min and strain.

BARISTA TIP The orange's caramelised sugar talks to the oolong's roast: sweetness without sugar.

10 Mid-afternoon Genmaicha

Toasted rice, comfort and little caffeine.

Hot · 4 min · 1 cup · 250 ml · Caffeinated · Easy

INGREDIENTS

4 g	genmaicha
250 ml	water at 85 °C

Finish. A few rice grains on the saucer.

METHOD

1. Steep for 1 min 30 s to 2 min.
2. Strain and drink. Do a second steep of 1 min.

BARISTA TIP Half the volume is rice: that is why its caffeine is lower than plain sencha.

11 Hojicha Latte

Toasted caramel for late afternoon.

Hot · 6 min · 1 mug · 250 ml · Caffeinated · With milk · Easy

INGREDIENTS

2 tbsp	hojicha leaves
100 ml	boiling water
150 ml	milk
1 tsp	honey

Finish. Foam and a pinch of cinnamon.

METHOD

1. Steep the hojicha in the water for 3 min (concentrate).
2. Heat the milk with the honey without boiling, and froth it.
3. Strain the hojicha into the mug and add the milk.

BARISTA TIP Roasting lowers the caffeine: the most sleep-friendly option that "tastes like coffee".

12 Green Tea & Coconut Water

Hydrates and wakes you up in the heat.

Iced · 5 min + 6 h · 1 bottle · 700 ml · Caffeinated · Easy

INGREDIENTS

6 g	green tea
400 ml	cold water
300 ml	chilled coconut water
½	lime, sliced

METHOD

1. Cold brew the green tea in the water for 6 h.
2. Strain and add the coconut water and lime.
3. Drink within 24 h.

BARISTA TIP Coconut water replaces potassium: a good choice for hot days or after exercise.

13 Iced Coconut Mate

Yerba mate and coconut, to take with you.

Iced · 5 min + 3 h · 1 bottle · 600 ml · Caffeinated · Easy

INGREDIENTS

2 tbsp	yerba mate
300 ml	cold water
300 ml	coconut water
1 tbsp	toasted grated coconut (optional)

METHOD

1. Combine the yerba and cold water in a bottle. Fridge for 3 h.
2. Strain through a paper filter.
3. Top up with the coconut water.

BARISTA TIP Coconut rounds off the yerba's bitterness. Shake before drinking.

14 Green Tea, Cacao & Chilli

Green tea with cacao nibs and a little heat.

Hot · 4 min · 1 cup · 250 ml · Caffeinated · Easy

INGREDIENTS

3 g	Chinese green tea
1 tsp	cacao nibs
1 pinch	chilli flakes or cayenne
250 ml	water at 80 °C

METHOD

1. Lightly crush the nibs.
2. Steep everything for 2 min 30 s.
3. Strain and drink unsweetened.

BARISTA TIP Cacao and chilli bring warmth and alertness without adding meaningful caffeine.

15 Ceylon with Cardamom and Honey

Bright black tea, fragrant, no milk.

Hot · 5 min · 1 mug · 250 ml · Caffeinated · Easy

INGREDIENTS

3 g	Ceylon black tea
2	crushed cardamom pods
250 ml	water at 95 °C
1 tsp	honey

METHOD

1. Steep the tea with the cardamom for 3 min.
2. Strain and sweeten with honey.

BARISTA TIP Ceylon is lighter and more citrusy than Assam: great on its own, without milk, mid-morning.

16 Concentration Kabusecha

Shaded green, rich in umami and L-theanine.

Hot · 5 min · 2 small cups · Caffeinated · Intermediate

INGREDIENTS

5 g	kabusecha or gyokuro
120 ml	water at 60 °C

METHOD

1. Cool the water by pouring it through two cups.
2. Steep 1 min 30 s in a kyusu or small pot.
3. Pour to the last drop. Repeat at 70 °C for 30 s.

BARISTA TIP Low water extracts the L-theanine and leaves caffeine and tannins behind: a broth-like taste and a calm mind.

17 Kukicha with Green Apple

Twig tea, almost caffeine-free, for the end of the day.

Hot · 5 min · 1 pot · 400 ml · Caffeinated · Easy

INGREDIENTS

4 g	kukicha (twig tea)
½	green apple, thinly sliced
400 ml	water at 85 °C

METHOD

1. Put the apple and kukicha in the pot.
2. Steep 2 min.
3. Strain. Make a second 3-minute steep.

BARISTA TIP Twigs have much less caffeine than leaves: ideal as the last cup before rooibos.

18 Rooibos & Mint: the Last Cup

Caffeine-free, to close the workday.

Hot · 6 min · 1 mug · 300 ml · Caffeine-free · Easy

INGREDIENTS

1 tbsp	rooibos
6 leaves	mint
300 ml	boiling water

METHOD

1. Steep the rooibos for 5 min.
2. Add the mint in the last minute.
3. Strain.

BARISTA TIP Set a time for this cup: it ends the workday. After it, no caffeine.

MORE RECIPES AND PLANNING

The weekly bottle

Sunday, 20 minutes: five days of tea ready in the fridge.

1. **Pick two teas**: one caffeinated for mornings (green or mate) and one caffeine-free for afternoons (rooibos, hibiscus or herbs).
2. **Make cold brew** in two 1 L jugs (8 to 12 g of leaf per litre).
3. **Strain on time** and move to clean bottles with lids.
4. **Label** with the name and date. Drink within 3 days; make the second batch on Wednesday.

DAY	MORNING	AFTERNOON
Monday	Green tea & coconut water (recipe 12)	Rooibos and orange
Tuesday	Iced coconut mate (recipe 13)	Hibiscus
Wednesday	Orange & rosemary tereré (recipe 6)	Mint and lemongrass
Thursday	Green tea with guaraná and lime (recipe 7)	Rooibos and orange
Friday	Green tea & coconut water (recipe 12)	Hibiscus

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Glass bottles with screw caps do not hold smells and show off the tea. Wash and scald between batches.

MORE RECIPES AND PLANNING

Snacks that help

Caffeine on an empty stomach spikes. A snack with fibre, good fat or protein smooths the curve.

TEA	SNACK	WHY IT WORKS
Matcha	Nuts and a square of 70% chocolate	Good fat, matching bitterness
Sencha	Edamame or onigiri	Umami with umami
Black tea	Wholegrain bread with cheese	Body and fibre
Mate	Banana with peanut butter	Slow energy
Oolong	Fresh fruit and yoghurt	Freshness and protein
Rooibos	Oat cookies	Light sweetness for day's end

Three simple rules

- Eat something in the first hour of the day, before or with the first cup.
- Drink a glass of water for every caffeinated cup.
- Choose low-sugar snacks: sugar gives the same spike you are trying to avoid.

> **NOTE**
> If you have anxiety, palpitations, gastritis or high blood pressure, talk to your doctor about your caffeine limit.



Thank you for brewing with us

The Botanora collection has ten guides, each devoted to one way of living with tea. Keep exploring:

01 Signature Infusions
02 The Tea Atlas
03 The Night Ritual
05 Iced & Cold Brew
06 Matcha at Home
07 Backyard Herbs
08 Crafting Blends
09 Tea & Table
10 Yerba Mate

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